American Computer Science League

2025-2026 ● Contest 2: Short Problems ● Classroom Division

1. Prefix/Infix/Postfix Notation

Convert the following infix expression to prefix:

$8 - 3 * 2 + 2^3 - 8 / 2$

A. − + − * ^ / 8 3 2 2 3 8 2
B. − − + 8 * 3 2 ^ 2 3 / 8 2
C. − + − 8 * 3 2 ^ 2 3 / 8 2
D. − + − 8 * 2 3 ^ 3 2 / 2 8
E. − + − 8 ^ 3 2 * 2 3 / 8 2

2. Prefix / Infix / Postfix

Evaluate the following postfix expression if a = 3, b = -2, and c = 5?

a c + a b − * a b − c + / 2 ^

A. $\frac{2}{5}$
B. 1
C. $\frac{16}{9}$
D. 16
E. -16

3. Bit-String Flicking

Evaluate:

NOT (10110 OR 11100 AND NOT 01101)

A. 10001
B. 00101
C. 00011
D. 10100
E. 01001

4. Bit-String Flicking

Evaluate:

(RCIRC-3 (LCIRC-2 01101) AND (RSHIFT-2 11010))

A. 01100
B. 10000
C. 00100
D. 10001
E. 01010

5. Prefix/Infix/Postfix Notation

Evaluate the following postfix expression if a = 3, b = -2, and c = 1:

a b * c + a 2 ^ − b c / +

A. -16
B. -14
C. 0
D. 14
E. 18

6. Prefix/Infix/Postfix Notation

Translate the following infix expression into prefix:

$$\frac{y^2z - z^2y}{y+z}$$

A. / − * ^ 2 y z * ^ 2 z y + z y
B. / − ^ y 2 z * * ^ z 2 y + y z
C. − / * ^ y 2 z * ^ z 2 y + y z
D. / − * ^ y 2 z * ^ z 2 y + y z
E. / − * ^ * ^ + y 2 z z 2 y y z

7. Bit-String Flicking

Evaluate:

(LCIRC-2 (RSHIFT-1 (NOT 01110))) OR (RCIRC-2 10011)

A. 11011
B. 11110
C. 11101
D. 10111
E. 01111

8. Bit-String Flicking

Solve the following equation for X if X is a 5-bit string:

(RCIRC-1 (LSHIFT-2 X AND 01101)) = 00010

A. *1*0*
B. ***01
C. *0*1*
D. ***10
E. 01***

9. LISP

Evaluate the following LISP expression:

(ADD (MULT 2 3) (SUB 9 3) (DIV 8 4))

A. 12
B. 13
C. 14
D. 15
E. 16

10. LISP

Evaluate the following LISP expression:

(CDR (CAR (REVERSE (CDR (CAR '((a b (c d)) (a b) c))))))

A. (b (c d))
B. c
C. d
D. (d)
E. (c d)

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